EX.NO: 3	Survey report on Emerging Trends in low-code application Development

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Abstract:

Recent research shows that low-code development is moving beyond simple drag-and-drop application building toward **AI-assisted development, citizen development, automation, cloud integration, governance, and enterprise-scale applications**. Gartner's 2025 forecast, for example, projects the low-code development technologies market to reach **$58.2 billion by 2029**, with AI agents and citizen development among the major adoption drivers.

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1. Introduction

Low-code application development is a modern software development approach that uses visual interfaces, drag-and-drop tools, reusable components, and minimal coding. It enables both developers and non-technical users to build applications faster. Modern low-code platforms also integrate AI, automation, data, and cloud services, making application development easier and more efficient.

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2.Overview of Low-Code Application Development

The evolution of application development can be represented as:

Traditional Coding → RAD → Low-Code → No-Code → AI-Assisted Low-Code → AI Agent-Based Development

Traditional Development

Developers manually write code using languages such as:

- Java
- Python
- C#
- JavaScript
- PHP

Low-Code Development

Developers use:

- Drag-and-drop interfaces
- Visual workflows
- Pre-built components
- Database connectors
- APIs
- Reusable templates

AI-Assisted Low-Code

The newest stage adds:

- Natural-language prompts
- AI-generated application components
- AI-assisted workflows
- Automated data processing
- AI agents
- Intelligent recommendations

Gartner's 2026 market analysis notes that leading low-code platforms are increasingly embedding **agentic AI directly into development environments**.

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3. Objectives and Research Methodology

Objectives

The main objectives of this survey are:

- To understand awareness of low-code development.
- To identify emerging trends in low-code technology.
- To study the benefits and challenges of low-code platforms.
- To understand the role of AI in low-code development.
- To analyze users' future interest in low-code platforms.

Research Methodology

The study uses a **survey-based research method**. Data can be collected using **Google Forms** from students, developers, faculty, and IT professionals. The collected responses are analyzed using **frequency, percentage, tables, charts, and graphs**.

4. Emerging Trends in Low-Code Development

Low-code development is evolving rapidly with new technologies.

Major emerging trends:

AI-Assisted Development: AI helps generate applications, workflows, formulas, and code from natural-language instructions.

Citizen Development: Non-programmers can create applications for their business requirements.

Workflow Automation: Repetitive tasks such as approvals, notifications, and data entry can be automated.

Cloud Integration: Applications can easily connect with cloud databases and services.

Data Analytics: Low-code platforms provide dashboards, reports, and data visualization.

AI Agents: Newer platforms are moving toward AI agents that can perform multi-step tasks with less manual configuration.

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5. Survey Questionnaire and Data Collection

A questionnaire is prepared to understand people's awareness, experience, opinions, and expectations regarding low-code development.

Sample questions:

1. Are you aware of low-code development?
2. Have you used a low-code platform?
3. Which low-code platform have you used?
4. What is the main benefit of low-code development?
5. Which emerging trend is most important?
6. What is the biggest challenge?
7. Do you think AI will increase low-code adoption?
8. Would you use low-code platforms in future?

The survey responses are collected through **Google Forms** and organized for further analysis.

6. Data Analysis and Statistical Interpretation

The collected survey data is analyzed to identify patterns and trends.

Percentage

Percentage is calculated using:

Percentage = (Number of responses / Total responses) × 100

Frequency

Frequency represents the **number of respondents** selecting a particular answer.

Mean

Mean can be used for questions based on a rating scale such as:

1 = Strongly Disagree → 5 = Strongly Agree

Example

If 70 out of 100 respondents are aware of low-code development:

Percentage = (70 / 100) × 100 = 70%

The results help identify the overall awareness, adoption, benefits, and concerns related to low-code development.

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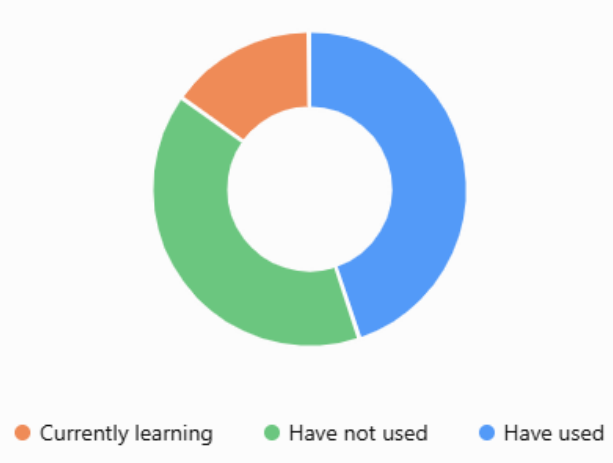
7. Charts and Graphical Representation

Charts make the survey results easier to understand and compare.

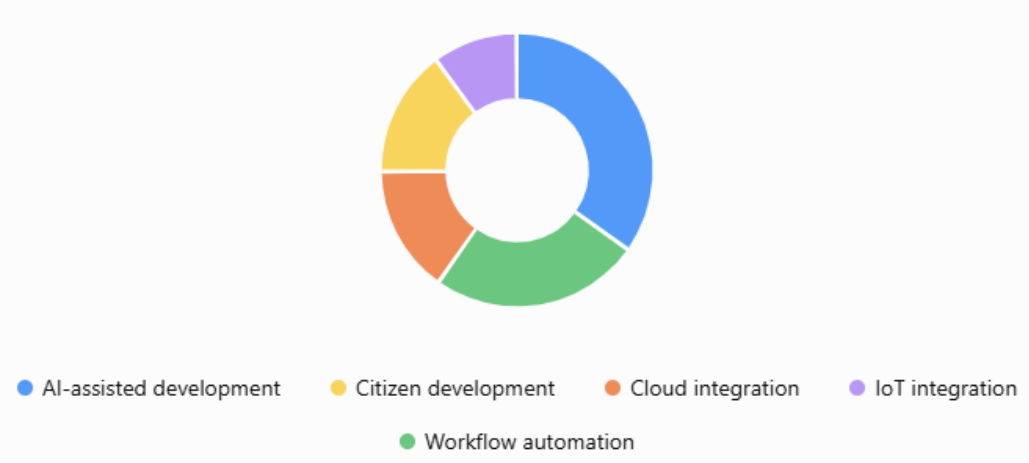
Awareness of Low-Code Development



Experience with Low-Code



Most Important Emerging Trend



8. Benefits and Challenges of Low-Code Development

Benefits

- Faster application development
- Reduced coding effort
- Lower development cost
- Easy prototyping
- Workflow automation
- Improved productivity
- Easier collaboration

- Greater participation from non-programmers

Challenges

- Security concerns
- Vendor lock-in
- Limited customization for complex applications
- Scalability issues
- Governance requirements
- Dependence on the platform
- Need for skilled developers for advanced applications

Therefore, low-code provides speed and simplicity but still requires proper planning and governance.

9. Real-World Applications and Future Trends

Low-code platforms are used in many areas such as:

- **Healthcare:** Patient management systems
- **Education:** Student management applications
- **Banking:** Internal workflow applications
- **Retail:** Inventory and sales management
- **Manufacturing:** Production monitoring
- **Human Resources:** Employee management
- **Business:** Customer and process management

Future Trends

The future of low-code development is expected to focus on:

AI + Low-Code + Cloud + Automation + Data Analytics + AI Agents

AI will increasingly help users create applications through natural-language instructions, while low-code platforms will continue to support faster business application development.

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10. Findings and Conclusion

Findings

The survey can be used to identify:

- The level of awareness of low-code development.
- The most commonly used platforms.
- The most important emerging trends.
- Major benefits expected by users.
- Main challenges affecting adoption.
- The expected impact of AI.
- Users' willingness to adopt low-code platforms in the future.

Conclusion

Low-code application development is becoming an important approach to modern software development. It reduces coding effort and enables faster application creation. The integration of **AI, automation, cloud computing, and data analytics** is making low-code platforms more powerful. However, challenges such as **security, vendor lock-in, scalability, and customization** must be considered. Low-code is therefore likely to **complement traditional programming rather than completely replace it**.

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